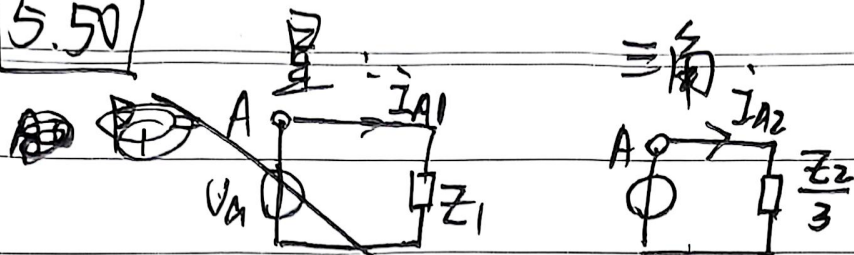


5.50



∴ 设 $U_{ab} = 100 \angle 0^\circ$, 则 $U_A = \frac{100}{\sqrt{3}} \angle -30^\circ$

则 $P_1 = 3 \operatorname{Re}[\dot{U}_A \dot{I}_{A1}]$

$\dot{I}_{A1} = \frac{\dot{U}_A}{Z_1}$

$\cos \varphi_1 =$

5.50 $Z_1: \dot{U}_A \dot{I}_{A1} Z_1 \therefore \dot{I}_{A1} = \frac{\dot{U}_A}{Z_1}$

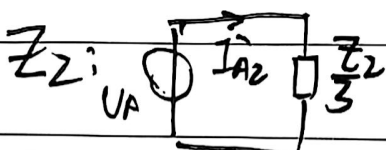
$P_1 = \operatorname{Re}[\dot{I}_{A1} \dot{U}_A^*]$

设 $U_{AB} = 100 \angle 0$, 则 $U_A = \frac{100}{\sqrt{3}} \angle -30^\circ$

∴ $P_1 = 3 \frac{|\dot{U}_A|^2}{|Z_1|} \cos \varphi_1 = 60 \text{ W} \therefore |Z_1| = \frac{27.78}{0.69} \Omega$

∴ $Z_1 = \frac{27.78}{0.69} \angle 60^\circ = 40.27 \angle 60^\circ$

∴ $\dot{I}_{A1} = \frac{\dot{U}_A}{Z_1} = \frac{100}{\sqrt{3} \cdot 40.27} \angle -90^\circ = 0.69 \angle -90^\circ$



$P_2 = 3 \frac{|\dot{U}_A|^2}{|Z_2|} \cos \varphi_2 = 45 \text{ W} \therefore |Z_2| = \frac{27.78}{0.43} \Omega$

∴ $Z_2 = \frac{27.78}{0.43} \angle 53^\circ = 64.6 \angle 53^\circ$

∴ $\dot{I}_{A2} = \frac{\dot{U}_A}{Z_2} = \frac{100}{\sqrt{3} \cdot 64.6} \angle 23^\circ = 0.43 \angle 23^\circ$

∴ $\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} = 0.05 - 1.12j + 0.396 - 0.522j = 0.655 \angle 52^\circ$

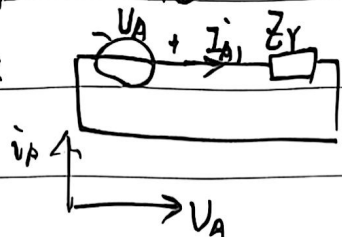
$I_B = 0.655 \angle -67.2^\circ$

$I_C = 0.655 \angle 172.8^\circ$

5.51

~~① Z_Y :~~ $\dot{V}_{AB} = 100\sqrt{3} \angle 30^\circ$ $\dot{U}_A = 100 \angle 0^\circ$

① Z_Y : $Q = 3\dot{U}_A \dot{I}_{A1} = 942 \text{ var.}$



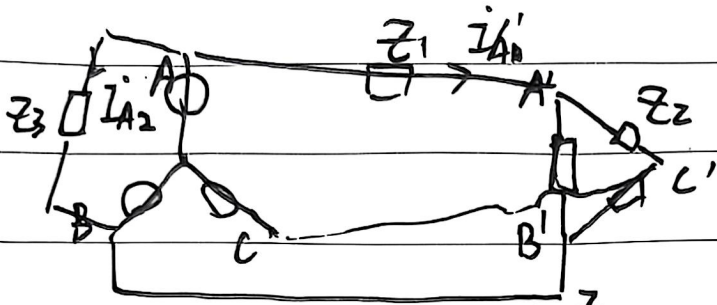
$\therefore \dot{I}_{A1} = \frac{942}{3.14} \angle 90^\circ \text{ A}$

② Z_Δ : $\dot{V}_{AB} = \dot{I}_{AB} Z_\Delta \Rightarrow \dot{I}_{AB} = \frac{\sqrt{3}}{3} \angle 30^\circ = 0.577 \angle 30^\circ \text{ A}$

$\dot{S} = 3[P + jQ]$ 其中 $P = \dot{V}_{AB}^* \dot{I}_{AB} = 100 \text{ W}$

$\therefore S = 300 + j \frac{942}{\sqrt{3}} \text{ W}$ 电抗

5.55 $|U_{AB}| = 380V$ $Z_1 = 2 + j4$ $Z_2 = 30 + j24$ $Z_3 = 38 \angle$



$$\dot{I}_A = \dot{I}_A' + \dot{I}_{A2}$$

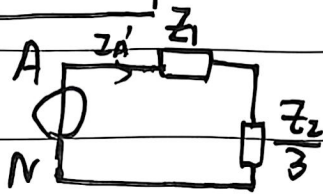
$$\dot{I}_B = \dot{I}_A' - \dot{I}_{A2}$$

$$\dot{I}_C = \dot{I}_A' \angle 120^\circ$$

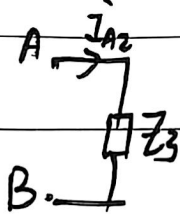
设 $\dot{U}_{AB} = 380 \angle 0^\circ$ 则 $\dot{U}_A = 220$

$$\dot{I}_A' = \frac{\dot{U}_A}{Z_1 + \frac{Z_2}{3}} = 12.96 \angle -75^\circ$$

下求 $\dot{I}_A'$:



② $\dot{I}_{A2}$



$$\dot{I}_{A2} = \frac{\dot{U}_{AB}}{Z_3} = 10 \angle -30^\circ$$

$$\therefore \dot{I}_A = 12.96 \angle -75^\circ + 10 \angle -30^\circ = 12.01 \angle -17.52^\circ$$

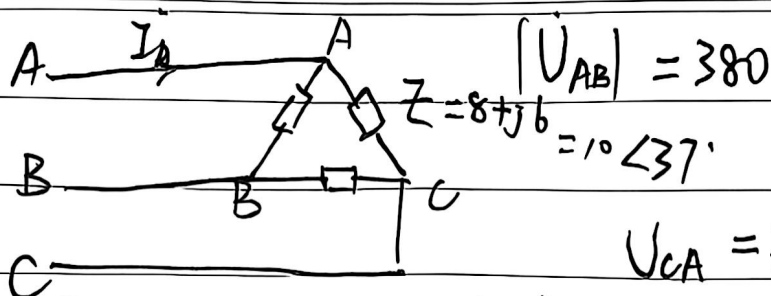
$$\dot{I}_B = 12.96 \angle -75^\circ - 10 \angle -30^\circ = -5.31 - j7.52$$

$$\dot{I}_C = 12.96 \angle 45^\circ$$

5.57

三相|S|相等, 但 P、Q 都不等 (有 cos φ)
 $\triangle: S = \dot{U} \dot{I}^* = -\dot{U}^* \dot{I}$

75240



$$U_{CA} = 380 \angle 150^\circ \quad U_{BC} = 380 \angle -90^\circ$$

(1) 设 $\dot{U}_{AB} = 380 \angle 0^\circ$, 则 $\dot{U}_A = 220 \angle -30^\circ$

① 线电流 $\dot{I}_A$

$$\dot{I}_A = \frac{\dot{U}_A}{\frac{Z}{3}} = 114 \angle -37^\circ$$

② $P_1 = \text{Re}[\dot{I}_A^* \dot{U}_{AB}] = 34656 \text{ W} = \text{Re}[25080 \angle 67^\circ] = 9799 \text{ W}$

③ 总有功功率 $S = 3 \dot{I}_A \dot{U}_{AB}^* = 69312 + j51984$ $P_2 = \text{Re}[\dot{I}_C^* \dot{U}_{CB}] = 24993 \text{ W}$
 $\therefore P_{\Sigma} = 34693 \text{ W}$
 $Q_{\Sigma} = 51984 \text{ W}$

(2) $P_1' = \text{Re}[\dot{I}_1^* \dot{U}_{AB}]$ $\dot{I}_1' = \frac{\dot{U}_{AB}}{Z} = 37.7 + j4.6$
 $\therefore P_1' = 11532 \text{ W}$

$P_2' = \text{Re}[\dot{U}_{CB} \dot{I}_C^*]$ $\dot{I}_C = \frac{\dot{U}_{CB}}{Z} = 38 \angle -53^\circ$

$\therefore P_2' = -11532 \text{ W}$